

Sungkyunkwan University

Department of Physics

**Measurement of HLT_IsoMu24
Trigger Efficiency
Using Tag-and-Probe Method
in Drell–Yan $Z \rightarrow \mu\mu$ Events**

with CMS Open Data at $\sqrt{s} = 13$ TeV

Siun Kim

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Advisor: Myeongjae Lee

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Abstract

The efficiency of the CMS high-level trigger path `HLT_IsoMu24` is measured using the Tag-and-Probe method in Drell–Yan $Z \rightarrow \mu\mu$ events. The measurement is performed using CMS Open Data from proton-proton collisions at $\sqrt{s} = 13$ TeV, corresponding to an integrated luminosity of 8.74 fb^{-1} recorded during the Run 2016H data-taking period. The trigger efficiency is evaluated as a function of the muon transverse momentum, pseudorapidity, and dimuon invariant mass by comparing the performance observed in data with that predicted by Monte Carlo simulation. Using the tight probe definition (CMS Tight ID and PF-based tight isolation), the data-to-simulation scale factor is measured to be 0.973 ± 0.001 (stat.) ± 0.001 (syst.), with a weak dependence on the kinematic variables. The Tag-and-Probe methodology and the measured scale factors are essential ingredients for precision measurements of Standard Model processes at the LHC.

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1 Introduction

The Standard Model (SM) of particle physics provides a remarkably successful description of the fundamental constituents of matter and their interactions. Among the key processes predicted by the SM, the Drell–Yan process [1], in which a quark-antiquark pair annihilates to produce a lepton pair via an intermediate Z boson or virtual photon ($q\bar{q} \rightarrow Z/\gamma^* \rightarrow \ell^+\ell^-$), serves as one of the most important benchmark processes at hadron colliders. Precision measurements of the Drell–Yan process provide stringent tests of perturbative quantum chromodynamics (QCD), constraints on parton distribution functions (PDFs), and a clean environment for detector calibration.

The Large Hadron Collider (LHC) at CERN collides proton beams at center-of-mass energies up to 13 TeV, producing vast numbers of Drell–Yan events. The Compact Muon Solenoid (CMS) experiment [2], one of the general-purpose detectors at the LHC, is designed to exploit this physics program with excellent muon reconstruction capabilities. However, the enormous collision rate at the LHC—approximately 40 MHz at the nominal bunch crossing frequency—far exceeds the rate at which events can be recorded for offline analysis. The CMS trigger system is therefore essential in selecting the most interesting events in real time, reducing the rate to a manageable level of approximately 1 kHz.

The trigger efficiency, defined as the fraction of offline-reconstructed events that also satisfy the online trigger requirements, must be precisely measured to enable accurate physics measurements. Any inefficiency or mismodeling of the trigger in Monte Carlo (MC) simulation directly propagates into systematic uncertainties on cross section measurements and other precision analyses. For muon-based analyses, the single-muon trigger path `HLT_IsoMu24`, which requires the presence of at least one isolated muon with transverse momentum $p_T > 24$ GeV at the high-level trigger (HLT), is one of the most widely used trigger paths at CMS.

The Tag-and-Probe (T&P) method [5] is the standard technique employed at CMS to measure trigger efficiencies in an unbiased manner using Drell–Yan events. In this method, one muon (the “tag”) is required to satisfy stringent identification and trigger criteria, ensuring that the event was recorded due to the tag muon. The second muon (the “probe”) is then used to measure the efficiency of interest, since its selection by the trigger is independent of the tag.

In this thesis, the efficiency of the `HLT_IsoMu24` trigger path is measured using the T&P method applied to $Z \rightarrow \mu\mu$ events. The analysis is performed using CMS Open Data [7] from proton-proton collisions at $\sqrt{s} = 13$ TeV, corresponding to the Run 2016H data-taking period. The trigger efficiency is measured as a function of the probe muon transverse momentum (p_T), pseudorapidity (η), and the dimuon invariant mass. The results from data are compared with MC simulation to extract data-to-simulation scale factors.

The structure of this thesis is as follows. Section 2 provides an overview of the CMS detector and its trigger system. Section 3 describes the data and MC samples used in this analysis, along with the event selection criteria. Section 4 details the Tag-and-Probe methodology. Section 5 presents the measured trigger efficiencies and scale factors. Finally, Section 6 summarizes the results and discusses their implications.

2 The CMS Detector and Trigger System

2.1 The Large Hadron Collider

The Large Hadron Collider (LHC) [3] is the world’s largest and most powerful particle accelerator, located at CERN near Geneva, Switzerland. It is housed in a circular tunnel with a circumference of approximately 27 km, previously used by the Large Electron-Positron Collider (LEP). The LHC accelerates two counter-rotating beams of protons to nearly the speed of light and brings them into collision at four interaction points around the ring, where the major experiments—ATLAS, CMS, ALICE, and LHCb—are located.

During the 2016 data-taking period (Run 2), the LHC operated at a center-of-mass energy of $\sqrt{s} = 13$ TeV with a bunch spacing of 25 ns, corresponding to a bunch crossing rate of 40 MHz. The peak instantaneous luminosity reached values exceeding $1.5 \times 10^{34} \text{ cm}^{-2}\text{s}^{-1}$, resulting in an average number of simultaneous proton-proton interactions (pileup) of approximately 23 per bunch crossing during Run 2016H.

2.2 The CMS Detector

The CMS detector [2] is a general-purpose detector designed to study a wide range of physics processes produced in high-energy proton-proton and heavy-ion collisions. The detector has a cylindrical geometry centered on the interaction point and covers nearly the full solid angle around the collision region. The central feature of the CMS apparatus is a superconducting solenoid of 6 m internal diameter, providing a magnetic field of 3.8 T.

The CMS coordinate system is defined with the origin at the nominal interaction point, the x -axis pointing toward the center of the LHC ring, the y -axis pointing vertically upward, and the z -axis along the counterclockwise beam direction. The azimuthal angle ϕ is measured in the x - y plane, and the polar angle θ is measured from the z -axis. The pseudorapidity is defined as $\eta = -\ln[\tan(\theta/2)]$.

The detector comprises four principal sub-detector systems, listed from innermost to outermost. The tracker and the calorimeters are located within the solenoid volume, while the muon system is embedded in the steel flux-return yoke outside the solenoid:

Silicon Tracker. The innermost detector is an all-silicon tracking system consisting of a pixel detector and a strip detector. The pixel detector, closest to the interaction point,

provides high-precision three-dimensional position measurements of charged particle trajectories. The strip detector extends the tracking coverage to larger radii. Together, they provide efficient tracking of charged particles with $|\eta| < 2.5$ and a transverse momentum resolution of approximately 1–2% for muons in the p_T range relevant to this analysis.

Electromagnetic Calorimeter (ECAL). Surrounding the tracker, the ECAL is a homogeneous calorimeter made of lead tungstate (PbWO_4) crystals, designed to measure the energy of electrons and photons with high precision and fine granularity.

Hadron Calorimeter (HCAL). The HCAL, located outside the ECAL, is a sampling calorimeter using brass absorber plates interleaved with plastic scintillator tiles. It measures the energy of hadrons and contributes to the measurement of missing transverse energy.

Muon System. The outermost component of CMS, and the most relevant for this analysis, is the muon detection system. It is embedded in the steel return yoke of the solenoid and consists of three types of gaseous detectors:

- **Drift Tubes (DT):** Located in the barrel region ($|\eta| < 1.2$), the DT chambers provide precise position measurements in the bending plane with a spatial resolution of approximately $100\ \mu\text{m}$ per chamber.
- **Cathode Strip Chambers (CSC):** Located in the endcap regions ($0.9 < |\eta| < 2.4$), the CSC chambers are multi-wire proportional chambers with cathode strip readout, capable of operating in the high-rate, non-uniform magnetic field environment of the endcaps.
- **Resistive Plate Chambers (RPC):** Located in both the barrel and endcap regions ($|\eta| < 1.9$), the RPCs provide fast timing information with a time resolution of about 1 ns, primarily used for triggering purposes.

The combination of the inner tracker and the muon system allows CMS to reconstruct muon tracks with excellent momentum resolution over a wide range of p_T and η , making the detector particularly well-suited for analyses involving muons.

2.3 The CMS Trigger System

The CMS trigger system [4] is a crucial component that reduces the 40 MHz bunch crossing rate to approximately 1 kHz, the maximum rate at which events can be written to permanent storage. The trigger system operates in two stages:

Level-1 (L1) Trigger. The L1 trigger is implemented in custom-designed hardware (FPGAs and ASICs) and processes information from the calorimeters and the muon system. It operates with a fixed latency of approximately $4\,\mu\text{s}$ and reduces the event rate from 40 MHz to about 100 kHz. For muon triggers, the L1 system uses information from the DT, CSC, and RPC detectors to identify muon candidates and estimate their p_{T} .

High-Level Trigger (HLT). The HLT is a software-based trigger system running on a large computing farm. It has access to the full detector readout and performs a more refined reconstruction of the event, similar to the offline reconstruction but with optimized algorithms for speed. The HLT reduces the rate from 100 kHz to approximately 1 kHz.

2.3.1 The HLT_IsoMu24 Trigger Path

The HLT_IsoMu24 trigger path is one of the primary single-muon trigger paths used by CMS during Run 2. It requires the presence of at least one muon candidate satisfying the following criteria at the HLT level:

- Transverse momentum $p_{\text{T}} > 24\,\text{GeV}$.
- The muon must be isolated, meaning that the sum of the transverse momenta of additional particles in a cone around the muon direction is below a certain threshold relative to the muon p_{T} . This isolation requirement suppresses backgrounds from muons produced within hadronic jets.

The efficiency of this trigger path—defined as the probability that a muon satisfying the offline reconstruction and identification criteria also passes the HLT_IsoMu24 requirements—is the primary quantity measured in this thesis.

3 Data and Monte Carlo Samples

3.1 CMS Open Data

This analysis uses CMS Open Data [7] released under CERN’s Open Data policy. The CMS collaboration has made a significant portion of its collision data publicly available, enabling independent physics analyses by researchers outside the collaboration. The data are provided in the NanoAOD format (version 9), a compact ROOT-based data format containing high-level physics objects suitable for analysis. The analysis is performed using the SKNanoAnalyzer framework [15], a C++ and ROOT-based [16] analysis tool developed by the Seoul National University CMS group for processing NanoAOD files. The full analysis code and documentation are publicly available [17].

3.2 Data Sample

The data sample used in this analysis consists of proton-proton collision events at $\sqrt{s} = 13$ TeV recorded by CMS during the Run 2016H data-taking period, corresponding to an integrated luminosity of 8.74 fb^{-1} . The events were collected using the `HLT_IsoMu24` trigger path and are contained in the `SingleMuon` primary dataset.

To ensure data quality, only events from luminosity sections certified as good for physics analysis are used. The certification is performed using the “Golden JSON” file, which lists the run numbers and luminosity sections that pass all data quality requirements.

3.3 Monte Carlo Simulation

For comparison with data, a simulated sample of Drell–Yan $Z \rightarrow \mu\mu$ events is used. The sample is generated using the POWHEG [8–10] next-to-leading-order (NLO) MC generator interfaced with PYTHIA 8 [11] for parton showering and hadronization, using the CP5 tune [12]. The simulated events include the full CMS detector simulation based on GEANT4 [13]. The cross section of the $Z \rightarrow \mu\mu$ process ($50 < m_{\mu\mu} < 120$ GeV) at NLO is $\sigma = 2116 \text{ pb}$.

The simulated events are reweighted to match the pileup distribution observed in data. In addition, the L1 prefiring correction is applied to the simulation to account for the mistiming of trigger primitives that affected the 2016 data-taking period. The Rochester correction [14] is applied to the muon momenta in both data and MC to account for residual misalignment and miscalibration effects in the muon p_T measurement. For data, the correction is a deterministic function of the muon kinematics. For MC, the correction uses the matched generator-level muon p_T when available, and a stochastic smearing otherwise.

Table 1 summarizes the data and MC samples used in this analysis. The MC sample contains 2 955 000 generated events, of which approximately 1% have negative generator weights (characteristic of NLO generators such as POWHEG), resulting in an effective number of events of approximately 2 897 000.

Table 1: Summary of data and Monte Carlo samples used in this analysis.

Sample	Generator	σ [pb]	Events	Eff. events
SingleMuon Run 2016H	(Data)	—	—	—
$Z \rightarrow \mu\mu$ (50–120 GeV)	POWHEG + PYTHIA 8	2116	2 955 000	2 897 000

3.4 Event Selection

The following event selection criteria are applied to both data and MC:

1. **Trigger:** Events are required to pass the HLT_IsoMu24 trigger. This is a prerequisite for the data to have been recorded, and the same requirement is applied to the MC for consistency.
2. **Muon multiplicity:** At least two muons are required in the event.
3. **Muon kinematics:** Muons are required to pass the CMS Loose identification with $p_T > 10$ GeV and $|\eta| < 2.4$ for the loose selection. For the tight selection, muons must additionally pass the CMS tight identification and the Particle-Flow (PF) based tight isolation criteria.
4. **Leading muon:** At least one muon with $p_T > 26$ GeV (the trigger-safe p_T cut, 2 GeV above the trigger threshold) is required.
5. **Opposite charge:** The tag and probe muons must have opposite electric charges.
6. **Mass window:** The invariant mass of the tag-probe pair must satisfy $|m_{\mu\mu} - 91.2 \text{ GeV}| < 10 \text{ GeV}$, corresponding to a window around the Z boson mass peak.

4 Tag-and-Probe Method

4.1 Overview

The Tag-and-Probe method is a data-driven technique widely used in high-energy physics to measure the efficiency of particle reconstruction, identification, and trigger algorithms. The method exploits the well-known properties of resonance decays—in this case, the decay of the Z boson into a pair of muons—to create an unbiased sample of probe particles whose efficiency can be measured.

The key idea is to select events containing a $Z \rightarrow \mu\mu$ candidate by requiring one muon (the “tag”) to satisfy stringent quality criteria, including matching to the trigger. The stringent tag requirement ensures that the event was recorded because of the tag muon, not the probe. The other muon (the “probe”) is then used to test whether it passes the selection criterion whose efficiency is being measured. Since the probe muon is selected independently of the trigger decision, the measurement is unbiased with respect to the trigger.

The efficiency ε is computed as:

$$\varepsilon = \frac{N_{\text{pass}}}{N_{\text{pass}} + N_{\text{fail}}} = \frac{N_{\text{pass}}}{N_{\text{all}}}, \quad (1)$$

where N_{pass} is the number of probes that pass the selection criterion and N_{all} is the total number of probes.

4.2 Tag and Probe Definitions

In this analysis, the Tag-and-Probe method is applied to measure the efficiency of the HLT_IsoMu24 trigger with respect to muons that have already passed the offline tight identification and isolation criteria. The definitions of the tag and probe muons are summarized in Table 2.

Table 2: Definition of tag and probe muons for the trigger efficiency measurement.

Tag muon	
Identification	CMS Tight ID
Isolation	PF-based tight relative isolation
Kinematics	$p_T > 26 \text{ GeV}$, $ \eta < 2.4$
Trigger matching	Matched to HLT trigger object ($\Delta R < 0.1$)
Probe muon (Tight)	
Identification	CMS Tight ID
Isolation	PF-based tight relative isolation
Kinematics	$p_T > 10 \text{ GeV}$, $ \eta < 2.4$
Probe muon (Loose)	
Identification	CMS Loose ID
Isolation	None
Kinematics	$p_T > 10 \text{ GeV}$, $ \eta < 2.4$
Tag-Probe pair	
Charge	Opposite sign
Mass window	$ m_{\mu\mu} - 91.2 \text{ GeV} < 10 \text{ GeV}$
Separation	$\Delta R(\text{tag, probe}) > 0.3$

The tag muon is the leading- p_T muon in the event that passes all the tight identification and isolation criteria and is additionally required to be geometrically matched to a trigger object. This matching is performed by requiring $\Delta R < 0.1$ between the offline muon and the nearest HLT muon trigger object, consistent with the CMS standard for HLT-level matching [6]. The requirement that the tag muon is trigger-matched ensures that the event was recorded because of the tag, making the probe muon's trigger decision independent.

Two probe definitions are used in this analysis:

- **Tight probe:** The probe muon is required to pass the CMS tight identification and tight PF-based isolation criteria. This definition measures the trigger efficiency *conditional* on the muon having already passed the offline ID and isolation requirements, i.e., $\varepsilon_{\text{Trigger|ID+Iso}}$. This is the standard definition used in CMS analyses [6].
- **Loose probe:** The probe muon is only required to pass the CMS Loose identification (with no isolation requirement) and minimal kinematic requirements ($p_T >$

10 GeV, $|\eta| < 2.4$). This definition provides a cross-check and measures a more inclusive trigger efficiency that includes contributions from muon identification and isolation effects.

4.3 Trigger Object Matching

Since the NanoAOD format does not store a pre-computed association between offline muons and trigger objects, the trigger matching is performed using the `TrigObj` collection stored in the NanoAOD files. For each offline muon, the matching algorithm searches for the nearest HLT trigger object satisfying:

- The trigger object is identified as a muon ($|\text{pdgId}| = 13$).
- The trigger object has `filterBits` bit 1 set, which for muon trigger objects indicates that the candidate passed the final isolation filter of the isolated single-muon HLT paths.
- The trigger object has $p_T > 24$ GeV (the `HLT_IsoMu24` threshold).
- The angular distance $\Delta R = \sqrt{(\Delta\eta)^2 + (\Delta\phi)^2} < 0.1$ between the offline muon and the trigger object.

The matching cone size of $\Delta R < 0.1$ was chosen following the CMS recommendation for HLT-level matching, where the tracker-based reconstruction provides sufficient angular resolution. This choice was validated by examining the ΔR distribution between tag muons and their nearest trigger objects, which shows a sharp peak at $\Delta R \approx 0.01$ – 0.03 with negligible entries beyond $\Delta R = 0.05$.

4.4 Efficiency Calculation

The trigger efficiency is measured as a function of three kinematic variables:

Efficiency vs. p_T : The efficiency is calculated in bins of probe muon p_T , providing the “turn-on curve” of the trigger. This is the most important distribution as it reveals the threshold behavior near $p_T = 24$ GeV.

Efficiency vs. η : The efficiency is calculated in bins of probe muon η for probes with $p_T > 26$ GeV (in the efficiency plateau region). This distribution reveals the geometric acceptance of the muon trigger system across the detector.

Efficiency vs. $m_{\mu\mu}$: The efficiency is calculated in 1 GeV bins of the tag-probe invariant mass in the range 86–96 GeV. This provides a cross-check of the measurement stability across the Z peak.

In this analysis, a simple counting method is employed rather than a fit-based approach. The efficiency in each bin is computed as:

$$\varepsilon = \frac{N_{\text{after trigger}}}{N_{\text{before trigger}}}, \quad (2)$$

where $N_{\text{before trigger}}$ and $N_{\text{after trigger}}$ are obtained from the integrals of the corresponding histograms. For the bin-by-bin efficiency distributions, the binomial division method is used, which properly accounts for the statistical uncertainties:

$$\sigma_\varepsilon = \sqrt{\frac{\varepsilon(1 - \varepsilon)}{N_{\text{before trigger}}}}. \quad (3)$$

The counting method is chosen over the fit-based approach commonly used in CMS Tag-and-Probe analyses for several reasons. First, the Z peak region is strongly dominated by signal events, and the background contamination is negligible (as demonstrated quantitatively in Section 5.2), making the signal extraction via fitting unnecessary. Second, the CMS Open Data environment does not provide access to the full set of fitting tools and templates used internally by CMS for signal-plus-background fits. Third, the counting method provides a transparent and reproducible measurement that is well-suited to the scope of this analysis.

4.5 Scale Factors

The data-to-simulation scale factor is defined as the ratio of the trigger efficiency measured in data to that predicted by the MC simulation:

$$\text{SF} = \frac{\varepsilon_{\text{Data}}}{\varepsilon_{\text{MC}}}. \quad (4)$$

A scale factor of unity indicates perfect agreement between data and simulation. Deviations from unity reflect residual differences in the trigger modeling that must be corrected when using MC simulation for physics measurements. In practice, the scale factor is applied as a per-event weight to the MC simulation to correct for trigger efficiency differences.

5 Results

5.1 Trigger Efficiency with Tight Probe Definition

Figure 1 shows the HLT_IsoMu24 trigger efficiency measured in data and MC simulation using the tight probe definition, as a function of the probe muon p_T (left), pseudorapidity η (center), and dimuon invariant mass (right). The upper panels display the efficiency and the lower panels show the data-to-MC scale factor.

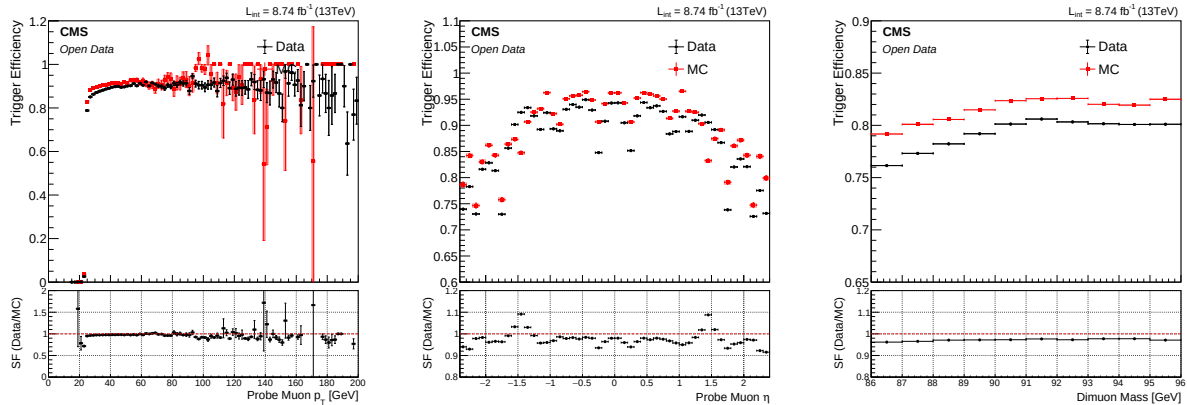


Figure 1: Trigger efficiency (upper panels) and data-to-MC scale factor (lower panels) using the tight probe definition, as a function of probe muon p_T (left), probe muon η for probes with $p_T > 26$ GeV (center), and dimuon invariant mass (right). Black points represent data and red points represent MC simulation. The dashed red line in the lower panels indicates $SF = 1$.

The p_T distribution (Figure 1, left) exhibits the expected turn-on behavior near the trigger threshold of 24 GeV, rising sharply from near zero at $p_T \approx 20$ GeV to a plateau above approximately 30 GeV. In the plateau region ($p_T > 30$ GeV), the data efficiency is approximately 87–91%, while the MC efficiency is approximately 90–93%. The resulting scale factor in the plateau region is approximately 0.97–0.98. At high p_T (> 80 GeV), the statistical uncertainties become large due to the limited number of probe muons, as the Z boson decay kinematics concentrate muon p_T around $m_Z/2 \approx 45$ GeV.

The η distribution (Figure 1, center) shows a clear dependence on pseudorapidity, reflecting the geometry of the CMS muon system. In the barrel region ($|\eta| < 1.2$), the efficiency is highest, reaching approximately 90–96% in MC and 85–95% in data, with localized dips near $\eta \approx \pm 0.25$ caused by the gaps between the wheels of the barrel muon chambers. The efficiency decreases in the endcap region, reaching values as low as approximately 73% in data around $|\eta| \approx 1.75$ and for $|\eta| > 2.1$. This behavior is consistent with the known features of the CMS muon trigger system [6]. The scale factor lies within approximately 0.92–0.99 for most bins, with localized excursions above unity (up to about 1.09) around $|\eta| \approx 1.4$ –1.5, where the simulation underestimates the efficiency in the barrel–endcap transition region. Apart from these localized features, the efficiency offset

between data and MC is largely independent of the detector geometry.

The mass-binned distribution (Figure 1, right) shows a mild increase in efficiency from the lower edge of the mass window (86 GeV) toward the Z peak (91 GeV), after which the efficiency plateaus. This is expected because events at the mass window edges tend to have more asymmetric muon p_T distributions. The scale factor is remarkably stable across the mass range, with values consistently around 0.96–0.98, indicating that the measurement does not depend sensitively on the choice of mass window.

5.2 Background Estimation

To validate the assumption that the background contamination is negligible in the Z peak region, the tag-probe invariant mass distribution is examined over a wide range of $60 < m_{\mu\mu} < 120$ GeV. Figure 2 shows the invariant mass distribution of all tag-probe pairs (before trigger requirement on the probe) and the subset where the probe passes the trigger (after trigger).

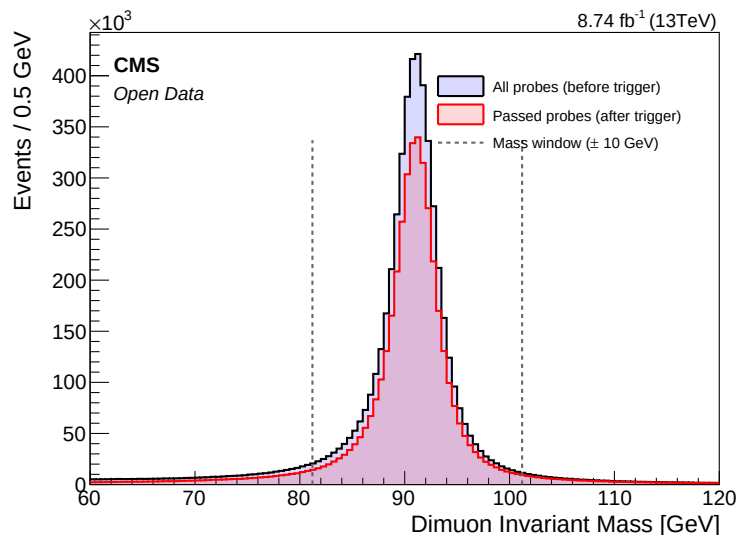


Figure 2: Tag-probe invariant mass distribution before (blue) and after (red) applying the trigger requirement on the probe muon, using the tight probe definition in data. The vertical dashed lines indicate the nominal mass window $|m_{\mu\mu} - 91.2| < 10$ GeV.

The distribution is sharply peaked at the Z boson mass of 91.2 GeV, with the vast majority of events concentrated within the nominal mass window of 81.2–101.2 GeV. The sideband regions (60–81.2 GeV and 101.2–120 GeV) contain approximately 11% as many events as the mass window. These sidebands are, however, dominated by genuine Drell–Yan events in the radiative tail of the Z resonance rather than by background: the Drell–Yan simulation alone, normalized to the data in the mass window, predicts a sideband-to-window ratio of 10.1%, compared with 10.8% observed in data. Attributing the small data excess over the simulated Drell–Yan shape in the sidebands to background, and extrapo-

lating this excess into the signal window, yields a background fraction of approximately 0.4%, i.e. below 1%. This high purity justifies the use of the simple counting method without the need for a signal-plus-background fit to extract the efficiency.

5.3 Comparison of Tight and Loose Probe Definitions

As a systematic cross-check, the trigger efficiency is also measured using the loose probe definition, in which the probe muon is only required to pass the CMS Loose identification ($p_T > 10$ GeV, $|\eta| < 2.4$) without any isolation requirement. This measurement provides insight into the importance of the probe definition and illustrates why the tight probe is the appropriate choice for the trigger efficiency measurement.

The results of the loose probe measurement are shown in Figure 3.

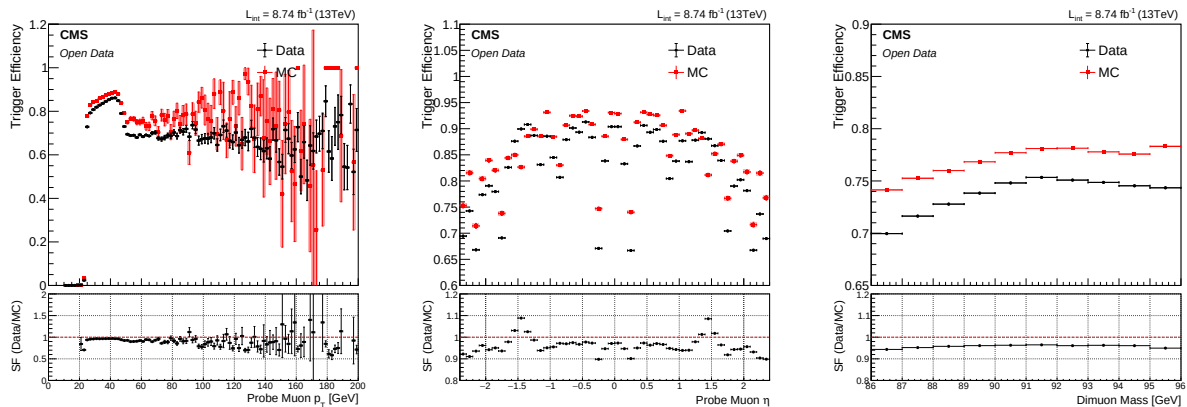


Figure 3: Trigger efficiency (upper panels) and data-to-MC scale factor (lower panels) using the **loose** probe definition, as a function of probe muon p_T (left), probe muon η (center), and dimuon invariant mass (right). The notation follows that of Figure 1.

The η distribution (Figure 3, center) exhibits a pattern qualitatively similar to the tight probe result, with scale factors mostly in the range 0.90–0.97, systematically slightly lower than the corresponding tight probe values, and with the same localized excursions above unity in the barrel–endcap transition region. The mass-binned efficiency (Figure 3, right) also shows a stable scale factor of approximately 0.94–0.96 across the Z peak, following the same pattern as the tight probe result.

However, the p_T distribution (Figure 3, left) reveals a striking difference from the tight probe result. While the tight probe shows the expected behavior—a sharp turn-on followed by a stable plateau—the loose probe efficiency reaches a maximum of about 86% in data at $p_T \approx 43$ GeV and then *decreases*, falling to a markedly lower level of approximately 70% for $p_T \gtrsim 50$ GeV, in both data and simulation. This behavior is unphysical for a trigger efficiency, which should remain at its plateau value once the threshold region is passed.

The origin of this effect is the contamination of the loose probe denominator by non-

isolated or poorly reconstructed muons. The loose probe definition applies only the CMS Loose identification with no isolation requirement, and at higher p_T the fraction of such muons that fail the tight ID and isolation criteria increases. These muons contribute to the denominator (N_{all}) but are unlikely to fire the isolation-based HLT_IsoMu24 trigger, artificially suppressing the measured efficiency. In contrast, the tight probe definition restricts the denominator to well-identified and isolated muons, which are precisely the muons that the trigger is designed to select.

This comparison demonstrates that the choice of probe definition has a significant impact on the measured efficiency and underscores the importance of matching the probe definition to the physics application. The trigger efficiency conditional on tight ID and isolation ($\varepsilon_{\text{Trigger}|\text{ID}+\text{Iso}}$), as measured with the tight probe, is the physically meaningful quantity for analyses that apply these offline selections. The loose probe results serve as a valuable cross-check and illustrate the robustness of the Tag-and-Probe methodology.

5.4 Systematic Uncertainties

Two sources of systematic uncertainty are evaluated: the choice of the invariant mass window used for the Tag-and-Probe selection, and the ΔR matching cone size used for trigger object matching. These variations are applied to the tight probe measurement, and the resulting change in the overall scale factor is taken as the systematic uncertainty.

Mass window variation. The nominal Tag-and-Probe selection requires the dimuon invariant mass to satisfy $|m_{\mu\mu} - 91.2 \text{ GeV}| < 10 \text{ GeV}$. To assess the sensitivity of the result to this choice, the half-width of the symmetric mass window is varied to 5, 15, and 20 GeV, and the overall efficiency is re-evaluated within each window. All windows, including the nominal $\pm 10 \text{ GeV}$, are evaluated with an identical procedure, so that the variations form a self-consistent family and the scale-factor differences are taken with respect to the $\pm 10 \text{ GeV}$ nominal. A wider window includes more events but also admits more background; a narrower window is purer but has less statistical power. The results are shown in Table 3.

Table 3: Overall trigger efficiency and scale factor for different mass window choices.

Mass window	$\varepsilon_{\text{Data}}$	ε_{MC}	SF	ΔSF
$\pm 5 \text{ GeV}$	0.7967	0.8188	0.9731	+0.0006
$\pm 10 \text{ GeV}$ (nominal)	0.7924	0.8148	0.9725	—
$\pm 15 \text{ GeV}$	0.7895	0.8122	0.9720	−0.0005
$\pm 20 \text{ GeV}$	0.7872	0.8103	0.9715	−0.0009

The scale factor varies by at most 0.0009 across the full range of mass window choices, indicating that the measurement is highly robust against this variation. The maximum

deviation, rounded to 0.001, is assigned as the systematic uncertainty from this source.

Trigger matching ΔR cone variation. The nominal matching cone of $\Delta R < 0.1$ is varied to 0.05, 0.15, and 0.2. The results are shown in Table 4.

Table 4: Overall trigger efficiency and scale factor for different ΔR matching cone sizes.

ΔR cone	$\varepsilon_{\text{Data}}$	ε_{MC}	SF	ΔSF
< 0.05	0.7897	0.8126	0.9718	-0.0006
< 0.10 (nominal)	0.7924	0.8148	0.9725	—
< 0.15	0.7931	0.8153	0.9728	+0.0003
< 0.20	0.7934	0.8155	0.9728	+0.0004

The scale factor is highly insensitive to the matching cone size, with variations below 0.001. This is consistent with the observation that the ΔR distribution between offline muons and trigger objects is sharply peaked at $\Delta R \approx 0.01$ – 0.03 , so that any cone size larger than approximately 0.05 captures essentially all true matches. The systematic uncertainty from this source is negligible compared with the mass-window variation.

Summary. Table 5 summarizes the systematic uncertainties on the overall scale factor.

Table 5: Summary of systematic uncertainties on the trigger efficiency scale factor.

Source	ΔSF
Mass window variation	0.001
ΔR matching cone size	< 0.001
Total systematic uncertainty	0.001

5.5 Summary of Scale Factors

Table 6 summarizes the trigger efficiency and scale factor obtained with both probe definitions, with the efficiency integrated inclusively over all probes with $p_T > 10$ GeV in the nominal mass window $|m_{\mu\mu} - 91.2 \text{ GeV}| < 10 \text{ GeV}$.

Table 6: Summary of the HLT_IsoMu24 trigger efficiency and data-to-MC scale factor for the tight and loose probe definitions. The efficiency is integrated inclusively over all probes with $p_T > 10$ GeV in the nominal mass window $|m_{\mu\mu} - 91.2 \text{ GeV}| < 10 \text{ GeV}$.

Probe definition	$\varepsilon_{\text{Data}}$	ε_{MC}	SF (Data/MC)
Tight (ID + Iso)	0.792 ± 0.001	0.815 ± 0.001	$0.973 \pm 0.001 \pm 0.001^*$
Loose (Loose ID)	0.735 ± 0.001	0.767 ± 0.001	0.958 ± 0.002

* First uncertainty is statistical, second is systematic.

The tight probe measurement yields a data efficiency of $(79.2 \pm 0.1)\%$ and a MC efficiency of $(81.5 \pm 0.1)\%$, resulting in a scale factor of 0.973 ± 0.001 (stat.) ± 0.001 (syst.). These efficiencies are integrated inclusively over all probe muons with $p_T > 10$ GeV and therefore include the trigger turn-on region; restricting the probes to the plateau ($p_T > 26$ GeV, the trigger-safe regime relevant for physics analyses) gives higher efficiencies of $(88.5 \pm 0.1)\%$ in data and $(90.6 \pm 0.1)\%$ in MC, with a consistent scale factor of 0.976. The loose probe yields a lower overall efficiency in both data and MC, as expected due to the inclusion of muons that pass only the Loose identification and fail the tight isolation criteria. The corresponding scale factor of 0.958 ± 0.002 is about 1.5% lower than the tight probe result, as expected, since mismodeling of the identification and isolation efficiencies also enters the loose probe measurement. The closeness of the two values nevertheless confirms that the bulk of the data-to-MC discrepancy originates from the trigger itself rather than from the probe selection.

5.6 Discussion

The measured scale factor of 0.973 is consistent with the range of 0.96–0.99 reported in the CMS muon trigger performance paper [6] for the HLT_IsoMu24 trigger during Run 2. The data efficiency lies about 3% below the simulation, and several factors may contribute to this residual difference from unity.

First, this analysis uses CMS Open Data, which does not provide access to the full suite of detector calibrations and corrections available to CMS collaboration members. The correction files used in this analysis (Rochester corrections, pileup reweighting) are taken from the publicly available 2016postVFP Ultra-Legacy calibrations, which were derived for the full 2016 post-VFP data-taking period (Runs F–H) rather than being optimized specifically for Run 2016H. Residual mismatches between the corrections and the specific running conditions of Run 2016H could lead to a systematic shift in the MC efficiency.

Second, the CMS official measurement employs a fit-based Tag-and-Probe method with detailed signal-plus-background modeling, whereas this analysis uses a simpler counting method. While the background is shown to be negligible (Section 5.2), subtle differences in the treatment of probe selection and event weighting may contribute to the small residual difference.

Third, the CMS official results are typically quoted as a function of p_T and η and averaged over the full Run 2 dataset, whereas this analysis covers only the Run 2016H period. Period-specific variations in detector performance, trigger configuration, or beam conditions could account for part of the remaining difference.

The measured scale factor is stable across all kinematic variables and consistent between the tight and loose probe definitions, and its good agreement with the official CMS

measurement indicates that the measurement methodology is sound. The primary value of this analysis lies in the demonstration of the Tag-and-Probe technique using publicly available data, rather than in replacing the official CMS calibration.

6 Conclusion

The efficiency of the CMS HLT_IsoMu24 trigger path has been measured using the Tag-and-Probe method in Drell–Yan $Z \rightarrow \mu\mu$ events from CMS Open Data collected during the Run 2016H data-taking period at $\sqrt{s} = 13$ TeV, corresponding to an integrated luminosity of 8.74 fb^{-1} .

The trigger efficiency has been evaluated as a function of the probe muon transverse momentum, pseudorapidity, and the dimuon invariant mass. The results demonstrate the expected features of the CMS single-muon trigger: a sharp turn-on at the trigger threshold of 24 GeV, a plateau efficiency of approximately 89% in data, a dependence on η reflecting the muon detector geometry, and stability across the Z mass peak.

The data-to-simulation scale factor has been measured to be 0.973 ± 0.001 (stat.) ± 0.001 (syst.) using the tight probe definition, in good agreement with the official CMS measurement. The systematic uncertainties, evaluated through variations of the mass window and the trigger matching cone size, are found to be small, confirming the robustness of the measurement.

A comparison between the tight and loose probe definitions reveals the critical importance of the probe selection. While the tight probe yields physically meaningful results with a stable efficiency plateau, the loose probe exhibits an anomalous decrease in efficiency at high p_T , attributed to the contamination of the denominator by non-isolated muons that pass only the Loose identification. This comparison validates the standard CMS practice of using the tight probe definition for trigger efficiency measurements.

The use of CMS Open Data demonstrates the viability of performing meaningful physics measurements with publicly available datasets, thereby promoting the principles of open science and reproducibility in high-energy physics. The Tag-and-Probe method, as applied in this analysis, provides a powerful and data-driven approach to measuring detector efficiencies, which is an essential ingredient for any precision measurement at the LHC.

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